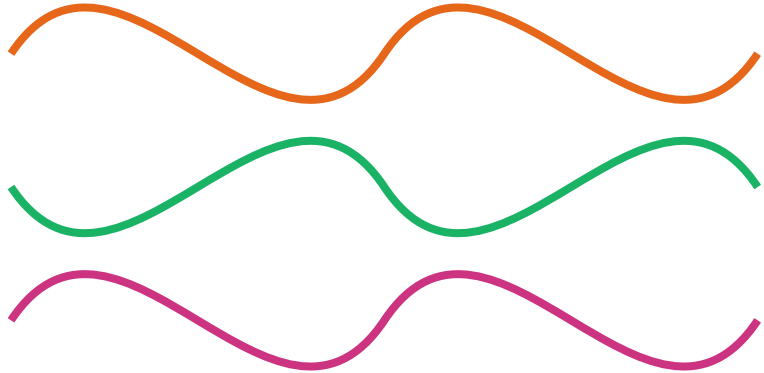


Page 1: Native Vector Pipeline

Rendered natively via DirectX and DirectWrite with infinite zoom sharpness



Vector Kerning Test

Exact glyph positioning from PDF font metrics & DirectWrite layout

ENGINE STATUS: 100% Native Vector

- o In Auto mode: Status bar displays lightning badge 'Vector'.
- o Visual sharpness: Zoom into this page up to 500% - lines and curves stay pin-sharp!
- o Zero raster artifacts: Geometry is tessellated directly on the GPU.

Page 2: Hybrid Fallback Pipeline

Automatic Fallback: Unsupported operators trigger Google PDFium with 0% truncation

NOTICE: This page contains an unsupported PDF operator - 'ri' / Rendering Intent

The Vector Engine classifies this construct during display list generation and gracefully delegates the entire page to Google PDFium.

Rendered via PDFium
100% Visual Fidelity

Zero Truncation
Safe Content Delivery

HOW TO VERIFY FALLBACK DIAGNOSTICS:

1. Look at the bottom-right status bar: it displays 'PDFium (Fallback)'.
2. Hover your mouse over the badge: the tooltip displays the exact reason:
'Unsupported PDF operator: ri'.
3. Switch to 'PDFium Only' in View -> Rendering Engine: renders via standard PDFium.
4. Switch to 'Vector Only' in View -> Rendering Engine: observes strict vector refusal.
5. Return to 'Auto': transparent fallback is restored instantly.

Page 3: Real-Time Engine Switching

Directly compare Windows ClearType vs FreeType antialiasing at all scales

7pt: The quick brown fox jumps over the lazy dog 1234567890 - Subpixel Font Hinting

9pt: The quick brown fox jumps over the lazy dog 1234567890 - Subpixel Font Hinting

11pt: The quick brown fox jumps over the lazy dog 1234567890 - Subpixel Font Hinting

14pt: Windows DirectWrite vs FreeType Software Rasterizer

20pt: Direct2D Hardware Acceleration

28pt: High-DPI Scalability



INTERACTIVE TEST EXPERIMENT:

1. Open the top menu: View -> Rendering Engine.
2. Switch between 'Auto' (Vector) and 'PDFium Only' (Raster).
3. Observe how the page instantly re-renders in real time.
4. Notice subtle differences in font smoothing, stroke weights, and GPU compositing.
5. Check the status bar: the badge updates dynamically to match the active engine!